Flent Lens — Complete Assumptions Register

Project: Flent Lens — Bengaluru Co-Living Arbitrage Analysis
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1. Flent's Business Model Assumptions

#	Assumption	Rationale
1.1	Flent operates a **rental arbitrage** model — it does NOT buy properties, it leases them.	The entire analysis is built around the monthly rent spread (lease cost vs. room revenue), not capital appreciation or purchase economics.
1.2	Flent leases large apartments (3BHK+) and	Each room in a 3BHK/4BHK becomes an independent

#	Assumption	Rationale
	converts them into **premium furnished co-living rooms**.	revenue unit, targeting young professionals and working tenants.
1.3	Flent charges tenants **per room**, not per apartment.	The arbitrage exists because per-room pricing for furnished co-living rooms exceeds a proportional share of the whole-apartment lease cost.
1.4	Flent's target city is **Bengaluru only**.	All data, ward boundaries (369 BBMP wards), and transit routes are Bengaluru-specific. The model is not generalised for other cities.
1.5	Three pillars determine ward viability: **Arbitrage Margin** (>₹5,000/month), **Supply Depth** (>3 listings of 3BHK+ per ward), and **Demand Intensity** (composite 0–1 index).	These form the minimum viability filter before a ward can even be scored.

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2. Sourcing & Acquisition Assumptions

#	Assumption	Rationale
2.1	Flent sources 3BHK/4BHK apartments at the **25th percentile (Q1) of ward-level rent**, not the median or average.	Flent targets distressed, bulk, or below-market inventory. Landlords willing to lease to an operator at scale accept lower rent in exchange for guaranteed occupancy, zero vacancy risk, and no tenant management. This is the "wholesale acquisition" thesis.

#	Assumption	Rationale
2.2	1BHK and 2BHK rents are aggregated at the **50th percentile (median)**.	These are the "retail" benchmark — the going rate a solo renter would pay for a standalone 1BHK. Flent's per-room pricing is anchored to this.
2.3	Only apartments with **≥1,100 sqft** are considered viable for 3BHK conversion.	Below 1,100 sqft, a 3BHK cannot physically yield 3 liveable, furnished rooms with adequate common space. This is the `MIN_SQFT_3BHK` config parameter.
2.4	A ward must have at least **3 listings per BHK type** to be considered statistically reliable.	Fewer than 3 listings makes median rent and quartile calculations statistically meaningless. Such wards are flagged `data_sparse = True`.
2.5	4BHK apartments are also viable acquisition targets — and may yield **higher margins** than 3BHKs if the space is large enough.	The model computes `arb_margin_best = max(arb_margin_3bhk, arb_margin_4bhk)`, always picking the more profitable conversion.

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3. Revenue Model Assumptions

#	Assumption	Rationale
3.1	Per-room revenue = **ward Median 1BHK Rent × 0.80 (Demand Discount Factor)**.	Tenants accept a 20% discount vs. renting a standalone 1BHK because co-living offers premium furnishing, community, zero deposit/maintenance hassle, and flexible lease terms.

#	Assumption	Rationale
3.2	The 0.80 Demand Discount Factor (DDF) is applied **uniformly** across all wards initially, then adjusted by an **elastic pricing multiplier** (±10%) based on local price pressure.	Premium wards where 1BHK rents exceed the city median can command a slightly higher DDF (up to 0.88), while lower-rent wards get a slightly lower one (down to 0.72). This avoids treating all wards identically.
3.3	Elastic pricing power is capped between **0.90× and 1.10×** of the base DDF.	This constrains the local adjustment so it never swings the DDF below 0.72 or above 0.88 — a realistic operating band.
3.4	Revenue is modelled on a **monthly** basis.	All margins, rents, and economic metrics are expressed as ₹/month. Annual or per-bed economics are not computed.
3.5	A ward's margin is considered **viable** only if the arbitrage margin percentage exceeds **5%** of the acquisition cost (`MARGIN_VIABLE_PCT = 0.05`).	A 5% spread is the minimum for covering operational costs (furnishing, maintenance, vacancy buffer). Below 5%, the ward is excluded regardless of absolute margin.

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4. Dynamic Yield / Room Conversion Assumptions

#	Assumption	Rationale
4.1	A **standard 3BHK** (<1,400 sqft) yields **3 revenue rooms**.	Each bedroom becomes one furnished co-living room.
4.2	A **large 3BHK** (1,400–1,600 sqft) yields **3.5 revenue rooms**.	Extra space allows conversion of a living room or servant quarter into a

#	Assumption	Rationale
		half-room (shared or premium).
4.3	An **XL 3BHK** (≥1,600 sqft) yields **4 revenue rooms**.	Sufficient space to partition the living area into a full additional room.
4.4	An **XL 4BHK** (≥2,000 sqft) yields **5 revenue rooms**.	4 bedrooms + 1 converted common area room.
4.5	A standard 4BHK (<2,000 sqft) yields **4 revenue rooms**.	No room for additional conversion beyond the existing bedrooms.
4.6	The "large" 3BHK threshold is **1,500 sqft** (for SFS size_adequacy) and the yield bracket threshold is **1,400 sqft** (for dynamic yield).	These two parameters serve different purposes — `LARGE_3BHK_SQFT` (1,500) gates the SFS bonus, while `YIELD_3BHK_LARGE_SQFT` (1,400) determines the 3.5-room yield.

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5. Demand Intensity Index (DII) Assumptions

#	Assumption	Rationale
5.1	DII is a **weighted composite** of three signals: Price Pressure (40%), Small Flat Concentration (30%), and PSF Spread (30%).	These three signals, taken together, approximate "how much unmet demand for affordable single-room housing exists in this ward."
5.2	**Price Pressure** = `ward_median_1bhk / city_median_1bhk`.	Wards where 1BHK rents are above the city median have upward price pressure — tenants are willing to pay, signalling strong demand.
5.3	**Small Flat Concentration (SFC)** = `(cnt_1bhk + cnt_2bhk) / total_listings`.	A ward saturated with 1BHK and 2BHK listings signals a market already oriented towards singles/couples — Flent's core demographic.

#	Assumption	Rationale
5.4	**PSF Spread** = `rps_1bhk - room_share_psf_3bhk`.	The per-sqft differential between renting a standalone 1BHK and the imputed per-sqft cost of one room-share in a 3BHK. A large positive spread means co-living can undercut solo 1BHK renting.
5.5	All three DII components are **min-max normalised** to [0, 1] before weighting.	Prevents any single metric from dominating the index due to scale differences.

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6. Supply Feasibility Score (SFS) Assumptions

#	Assumption	Rationale
6.1	SFS is a **weighted composite** of: Supply Volume (35%), Size Adequacy (30%), and Capital Efficiency / ROI (35%).	Measures whether a ward has enough convertible stock, of adequate size, with good return on capital.
6.2	**Supply Volume** = `log1p(cnt_3bhk + cnt_4bhk) / log1p(city_max)`.	Log-scaled to prevent wards with massive listing counts from overpowering the index. Measures relative supply depth.
6.3	**Size Adequacy** = percentage of 3BHK listings in a ward that are ≥1,500 sqft.	Larger apartments yield more rooms per lease — higher capital efficiency for Flent.
6.4	**Capital Efficiency** = `arb_margin_best / avg_rent_3bhk`.	ROI per unit of capital deployed. High absolute margin with low lease cost = maximum efficiency.
6.5	A **size bonus** multiplier (0.8–1.2×) is applied based	Rewards wards where average 3BHK size naturally

#	Assumption	Rationale
	on `avg_sqft_3bhk / 1,500`.	tends larger, and penalises cramped wards.

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7. Effective Margin Density (EMD) Assumptions

#	Assumption	Rationale
7.1	EMD = `arb_margin_best × (cnt_3bhk + cnt_4bhk) / area_sqkm`.	Measures total addressable arbitrage potential per square kilometre. A ward with ₹10K margin but only 2 listings has low EMD; a ward with ₹8K margin and 20 listings has high EMD.
7.2	If ward area is unavailable, EMD falls back to `arb_margin_best / avg_sqft_3bhk × 1000`.	A simplified proxy when spatial area data is missing.
7.3	EMD is a **diagnostic metric**, not directly used in the composite score.	It informs operational decisions (where to deploy sourcing teams) but the Opportunity Score uses normalised components instead.

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8. Transit Score Assumptions

#	Assumption	Rationale
8.1	Transit is treated as a **penalty** (negative weight), not a boost.	The original model weighted transit positively. However, high-transit wards tend to be congested, noisy, and less desirable for premium co-living. Transit is now subtracted from the

#	Assumption	Rationale
		opportunity score with weight −0.10.
8.2	Transit score is a weighted average of **route density** (50%) and **route length density** (50%).	Route density = number of BMTC routes / ward area (km²). Length density = total route length / (ward area × 1000).
8.3	Both sub-scores are **percentile-ranked** before combining.	Uses `rank(pct=True)` to transform raw values to a [0, 1] scale based on relative position.
8.4	Only **BMTC bus routes** are used as the transit signal.	Metro, rail, and auto-rickshaw data are not included. This is a limitation — transit score only reflects bus connectivity.
8.5	Routes are **clipped to ward boundaries** via geometric intersection before counting.	A bus route passing through a ward for 200m counts differently than one traversing 5km through it.

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9. SEZ Employment Score Assumptions

#	Assumption	Rationale
9.1	Proximity to SEZs (tech parks) is a **strong demand driver** for co-living.	SEZ employees are Flent's target demographic — young professionals who prefer furnished rooms near work.
9.2	SEZ score uses a **gravity model**: `score = Σ (1 / distance^α)` summed over all SEZ zones.	Closer wards get exponentially higher scores. Wards near multiple SEZs benefit cumulatively.
9.3	The distance decay exponent `α = 1.5`.	Balances between linear decay (α=1, too flat) and inverse-square (α=2, too

#	Assumption	Rationale
		aggressive). At α=1.5, a ward 2km from an SEZ scores ~3.5× higher than one at 5km.
9.4	A **minimum distance floor** of 500m is applied.	Prevents divide-by-zero and avoids unrealistically inflated scores for wards whose centroid happens to sit inside an SEZ polygon.
9.5	Distance is measured from **ward centroid** to SEZ point.	A simplification — actual distance from the ward boundary would be more precise, but centroid-based distance is computationally efficient and sufficiently accurate at the ward-level scale.
9.6	SEZ locations are **geocoded via Nominatim** (OpenStreetMap) from developer names and locations.	Some SEZs may geocode imprecisely. Results are cached to avoid re-geocoding.
9.7	Each SEZ is represented as a **25-acre circle** (radius ≈ 179.45m) in the KML overlay.	This is a mathematical approximation — real SEZ boundaries are irregular. 25 acres ≈ 101,171 m², giving radius = √(101171/π) ≈ 179.45m.
9.8	SEZ employment score has a **positive weight of +0.20** in the composite score.	SEZ proximity is a strong demand signal and is weighted higher than transit.

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10. Composite Opportunity Score Assumptions

#	Assumption	Rationale
10.1	The composite score formula is: `OPP_SCORE =`	Arbitrage margin is the primary signal (45%),

#	Assumption	Rationale
	`(ARB × 0.45) + (DII × 0.30) + (SFS × 0.25) + (SEZ × 0.20) - (TRANSIT × 0.10)`	followed by demand (30%), supply (25%), SEZ proximity (20% boost), and transit (10% penalty).
10.2	The raw score is **multiplied by 100** for readability.	Converts the 0–1 weighted sum into a 0–100+ scale.
10.3	All component inputs are **min-max normalised** to [0, 1] before weighting.	Ensures no single metric's raw scale dominates the composite.
10.4	The economic weights (ARB + DII + SFS) sum to **1.00**. The overlays (SEZ, Transit) are additive/subtractive on top.	This means the final score can technically exceed 100 (if overlays are strong) or go negative (if transit penalty is large on a low-ECON ward).
10.5	After spatial spillover adjustments, the score is **re-indexed to 100** based on the best data-rich, viable ward.	The top realistic ward always scores 100, making tier thresholds interpretable.

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11. Spatial Spillover / Aura & Island Assumptions

#	Assumption	Rationale
11.1	**Aura = Neighborhood Contagion.** A ward surrounded by strong neighbors is more investable than an identical ward in isolation.	Co-living demand spills across ward boundaries. A cluster of Tier 1 wards creates operational density for Flent (shared maintenance teams, brand visibility).
11.2	Spatial adjacency is determined by **Queen contiguity** — wards that share any boundary point (even a corner) are neighbors.	Queen contiguity is more inclusive than Rook (sides only), capturing the real-world fact that renters don't distinguish between sharing a long border vs. a corner.

#	Assumption	Rationale
11.3	**Tier 1 neighbor boost: +10%** (`OPP_SCORE × 1.10`).	If your best neighbor is Tier 1, you get the maximum aura boost.
11.4	**Tier 2 neighbor boost: +5%** (`OPP_SCORE × 1.05`).	If your best neighbor is Tier 2 (and no Tier 1 neighbor), a moderate boost.
11.5	**Tier 3 neighbor boost: +2.5%** (`OPP_SCORE × 1.025`).	Minimal boost — the neighbor is viable but not strong.
11.6	**Island Penalty: −15%** (`OPP_SCORE × 0.85`) for Tier 1 wards with **zero** Tier 1 or Tier 2 neighbors.	A Tier 1 ward surrounded only by Tier 3/Excluded wards is a spatial outlier — likely a data artifact or a non-scalable pocket. Penalised to avoid recommending isolated investments.
11.7	A ward receives the boost from its **highest-tier neighbor only** (not cumulative).	This is a "best neighbor wins" model, not an average of all neighbors. Prevents score inflation in densely connected areas.
11.8	Spatial spillover is applied **before** final tier assignment but **after** the base Opportunity Score is computed.	The base score captures economics; spillover then adjusts for geographic context.

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12. Tier Assignment Assumptions

#	Assumption	Rationale
12.1	Tiers are assigned on **percentile bands of eligible wards only** — not all 369 wards.	Eligible = `margin_viable AND NOT data_sparse`. Using only eligible wards for thresholds prevents

#	Assumption	Rationale
		excluded/sparse wards from skewing the cutoffs.
12.2	**Tier 1 = ≥75th percentile** — Priority sourcing targets.	Top quartile of viable wards. These are the wards where Flent should deploy sourcing teams immediately.
12.3	**Tier 2 = ≥50th percentile** — Secondary pipeline.	Above-median wards. Worth monitoring and converting when market conditions improve or supply opens up.
12.4	**Tier 3 = ≥25th percentile** — Watch list.	Viable but not compelling enough for immediate investment. May improve with infrastructure changes.
12.5	**Excluded = <25th percentile OR margin not viable.**	Not recommended. Either the economics don't work or there's insufficient data.
12.6	**Data-sparse wards are capped at Tier 3** even if their OPP_SCORE places them in Tier 1 or 2.	A ward with 2 listings might show amazing margins, but the data is unreliable. Conservative treatment prevents false positives.
12.7	The percentile thresholds (75/50/25) are **configurable** via the dashboard in real-time.	Leadership can tighten (e.g., 80/60/40) to be more selective or loosen to expand the target list.

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13. Data Quality & Cleaning Assumptions

#	Assumption	Rationale
13.1	**Only apartment-type properties** are included: Apartment, Multistorey Apartment, Builder Floor Apartment, Penthouse, Studio Apartment.	Villas, plots, PGs, independent houses, and commercial properties are excluded as they don't fit the co-living conversion model.

#	Assumption	Rationale
13.2	**All listings are rental** (`listing_type = 'rent'`).	Sale listings are excluded — Flent leases, it doesn't buy.
13.3	Monthly rent must be between **₹5,000 and ₹5,00,000**.	Below ₹5,000 is likely erroneous or a parking space. Above ₹5,00,000 is ultra-luxury and not target inventory.
13.4	BHK type must be between **1 and 6**.	Higher values are likely data entry errors.
13.5	1BHK listings with rent **>₹50,000 are dropped** as anomalies.	A 1BHK at ₹50K+ is either mislabelled or an ultra-luxury outlier that would inflate the city median and distort arbitrage calculations.
13.6	Outliers are removed per **ward-BHK group** using a **3σ (standard deviation) threshold**.	Any listing whose rent deviates more than 3 standard deviations from its ward-BHK group median is considered a statistical outlier and is dropped.
13.7	**Geo-fence:** Only listings within lat 12.7°–13.2° and lon 77.4°–77.8° are accepted.	This bounding box covers Greater Bengaluru. Listings outside it are either geocoding errors or outside the BBMP jurisdiction.
13.8	Listings with **zero or near-zero coordinates** (	lat
13.9	BHK type is extracted via **regex heuristics** from: (1) the `bhk_type` field, (2) the listing URL, (3) the apartment bio/description.	Magicbricks data has inconsistent BHK labelling. The regex cascade recovers ~90%+ of missing BHK values.
13.10	Studio apartments are treated as **1BHK equivalents**.	For rent comparison purposes, a studio and a 1BHK serve the same market.

#	Assumption	Rationale
13.11	Deduplication is performed on both `listing_id` and `listing_url`.	The same property may appear with different IDs, and the same listing may be duplicated with different URLs. Both are removed.

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14. Spatial & Geographic Assumptions

#	Assumption	Rationale
14.1	**369 BBMP wards** (December 2025 boundary definitions) are used as the geographic unit of analysis.	This is the latest available ward boundary KML from the Greater Bengaluru Authority (GBA).
14.2	Listings are assigned to wards via **point-in-polygon spatial join** (primary method).	A listing's lat/lon coordinate is tested against all 369 ward polygons.
14.3	Listings that fall outside all ward polygons get a **proximity fallback** — assigned to the nearest ward within **2,450 metres**.	Some listings geocode to roads, intersections, or slightly outside ward boundaries. The 2,450m max distance prevents assigning listings to wards they're clearly not in.
14.4	Listings **>2,450m from any ward boundary** are treated as truly outside BBMP and are dropped.	These are either geocoding errors or properties genuinely outside the city.
14.5	Ward areas are computed by **projecting to UTM Zone 43N** (EPSG:32643), calculating area, then converting back to WGS84.	UTM provides accurate area measurements in square metres, unlike WGS84 which uses degrees.
14.6	Ward IDs are extracted from the KML's **globally unique `id` field** (e.g.,	The raw `ward_id` field in the KML is only unique within each zone (1–N per zone), causing collisions

#	Assumption	Rationale
	`ward_369_final.123` → `123`).	when wards from different zones share the same number. The global `id` prevents this.

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15. Econometric Validation Assumptions

#	Assumption	Rationale
15.1	**Moran's I** is used to test whether rental prices are spatially autocorrelated (clustered).	A positive Moran's I (>0.3, p<0.05) confirms that rent patterns are geographically structured — not random — validating the spatial spillover model.
15.2	The spatial weights matrix for Moran's I uses **Queen contiguity** with `silence_warnings=True`.	Queen contiguity = wards sharing any boundary point. `silence_warnings=True` suppresses warnings about island wards (wards with no contiguous neighbors, which can exist at the BBMP boundary edge).
15.3	If Moran's I p-value ≥ 0.05, the spatial clustering is **not statistically significant**, but the model still runs.	The validation informs confidence level — it doesn't gate the analysis.
15.4	**OLS Regression: Q25_3BHK_Rent ~ Median_1BHK_Rent** validates the structural profitability of the arbitrage model.	If the slope (β) divided by 3 is less than the operational DDF (0.80), arbitrage is structurally profitable at the market level.
15.5	OLS uses **HC3 heteroscedasticity-robust standard errors**.	Rental data is heteroscedastic (variance increases with rent level). HC3 corrects for this, preventing false significance.

#	Assumption	Rationale
15.6	The **breakeven demand discount factor** = β / 3 (assuming 3 rooms from a 3BHK).	If Flent's operational DDF (0.80) exceeds this breakeven, the arbitrage is structurally valid across the market.
15.7	A minimum of **5 data points** is required for OLS regression to run.	Fewer than 5 ward-level observations do not permit meaningful regression.

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16. Dashboard & Configuration Assumptions

#	Assumption	Rationale
16.1	The dashboard **never modifies `config.py`**. All parameter changes are held in session memory only.	Prevents accidental overwriting of the validated baseline. A server restart atomically resets all parameters.
16.2	Any parameter change in the dashboard triggers an **automatic pipeline re-run** after a 3-second debounce.	The debounce prevents the pipeline from firing on every slider tick during a continuous drag.
16.3	The pipeline subprocess receives overridden parameters via a **temporary JSON file** that is deleted after execution.	Clean handoff — no shared memory between Flask and the pipeline subprocess.
16.4	All **weights, thresholds, and percentile cutoffs** are configurable in real-time.	Leadership can explore "what if" scenarios without engineering support.

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17. Data Source Assumptions

#	Assumption	Rationale
17.1	**Magicbricks** is the sole source of rental listing data.	No data from 99acres, NoBroker, Housing.com, or other platforms. Magicbricks was chosen for its API accessibility and listing density in Bengaluru.
17.2	The data represents a **point-in-time snapshot**, not a time-series.	Rent trends, seasonality, and market velocity are not captured. The analysis assumes current listings reflect stable market conditions.
17.3	**BMTC bus routes** (KML) represent the **current bus network** at the time of data collection.	Future route changes, metro expansions, and suburban rail are not modelled.
17.4	**SEZ locations** are from the official government SEZ developer list.	Some SEZs may be denotified, under construction, or have changed names. Geocoding via Nominatim may not resolve all names accurately.
17.5	The **BBMP ward boundaries** (December 2025 KML) reflect the most recent delimitation.	Ward boundaries may change with future revisions. The analysis is locked to this specific boundary set.

Summary of Key Numerical Assumptions

Parameter	Value	Meaning
`DEMAND_DISCOUNT_FACTOR`	**0.80**	Tenants pay 80% of standalone 1BHK rent for a co-living room

Parameter	Value	Meaning
`MARGIN_VIABLE_PCT`	**5%**	Minimum arbitrage margin (as % of acquisition cost) for viability
`MIN_SQFT_3BHK`	**1,100 sqft**	Minimum flat size for 3BHK co-living conversion
`LARGE_3BHK_SQFT`	**1,500 sqft**	Size adequacy threshold for SFS scoring
`OUTLIER_STD_THRESHOLD`	**3.0σ**	Outlier removal — listings beyond 3 standard deviations
`MAX_VALID_1BHK_RENT`	**₹50,000**	Cap on 1BHK rent to filter anomalies
`MIN_LISTINGS_PER_BHK`	**3**	Minimum listings for statistical reliability
`NEAREST_WARD_MAX_DISTANCE_M`	**2,450m**	Max distance for proximity-based ward assignment
`SEZ_DISTANCE_DECAY_ALPHA`	**1.5**	Gravity model exponent for SEZ proximity
`SEZ_MIN_DISTANCE_M`	**500m**	Floor distance to prevent divide-by-zero
`ECON_WEIGHT_ARBITRAGE`	**0.45**	Weight of arbitrage margin in composite score
`ECON_WEIGHT_DII`	**0.30**	Weight of demand intensity in composite score
`ECON_WEIGHT_SFS`	**0.25**	Weight of supply feasibility in composite score
`OVERLAY_WEIGHT_TRANSIT`	**−0.10**	Transit penalty weight (subtract from score)
`OVERLAY_WEIGHT_SEZ`	**+0.20**	SEZ proximity boost weight
`SPILLOVER_BOOST_TIER1`	**+10%**	Aura boost for Tier 1 neighbors

Parameter	Value	Meaning
`SPILLOVER_BOOST_TIER2`	**+5%**	Aura boost for Tier 2 neighbors
`SPILLOVER_BOOST_TIER3`	**+2.5%**	Aura boost for Tier 3 neighbors
`ISOLATION_PENALTY`	**−15%**	Island penalty for isolated Tier 1 wards
`TIER1_PERCENTILE`	**75th**	Top quartile = Tier 1
`TIER2_PERCENTILE`	**50th**	Above median = Tier 2
`TIER3_PERCENTILE`	**25th**	Above 25th percentile = Tier 3
`TRANSIT_WEIGHT_DENSITY`	**0.50**	Weight of route count density in transit score
`TRANSIT_WEIGHT_LENGTH`	**0.50**	Weight of route length density in transit score
Yield: 3BHK <1,400 sqft	**3 rooms**	Standard conversion
Yield: 3BHK 1,400–1,600 sqft	**3.5 rooms**	Large conversion
Yield: 3BHK ≥1,600 sqft	**4 rooms**	XL conversion
Yield: 4BHK ≥2,000 sqft	**5 rooms**	XL 4BHK conversion

Limitations & Caveats

1. **Single data source** — Magicbricks only. Cross-platform validation would improve robustness.
2. **No time-series** — Rent trends and vacancy rates are not captured.
3. **Transit = bus only** — Metro and suburban rail are not modelled.
4. **SEZ geocoding accuracy** — Nominatim may resolve some SEZ names imprecisely.
5. **Ward centroid simplification** — SEZ distances are measured from ward centroids, not boundaries.
6. **Static demand discount** — The 0.80 DDF is assumed uniform (with ±10% local elastic band), but actual co-living pricing may vary by property, furnishing quality, and brand positioning.
7. **No capex modelling** — Furnishing costs, security deposits, and setup capital are not included in the margin calculation.

8. **No vacancy modelling** — The model assumes 100% occupancy. Real-world vacancy rates would reduce effective margins.

This document captures every assumption embedded in the Flent Lens codebase as of April 2026. All assumptions are configurable via `config.py` or the Leadership Dashboard.